

IMO(25th) – 1984

First Day

1. Let x, y, z be non negative real numbers with $x + y + z = 1$. Show that $0 \leq xy + yz + zx - 2xyz \leq \frac{7}{27}$.
2. Find two positive integers f such that non of the numbers $a, b, a + b$ is divisible by 7 and $(a + b)^7 - a^7 - b^7$ is divisible by 7^7 .
3. In a plane two different points O and A are given. For each point $X \neq O$ of the plane denote by $\alpha(X)$ the angle AOX measured in radians ($0 \leq \alpha(X) < 2\pi$) and by $C(X)$ the circle with centre O and radius $OX + \frac{\alpha(X)}{OX}$. Suppose each point of the plane is colored by one of the finite number of colors. Show that there exists a point X with $\alpha(X) > 0$ such that its color appears somewhere on the circle $C(X)$.

Second Day

4. Let $ABCD$ be a convex quadrilateral for which the circle of diameter AB is tangent to the line CD . Show that the circle of diameter CD is tangent to the line AB if and only if the lines BC and AD are parallel.
5. Let d be the sum of the lengths of all diagonals of a convex polygon of n ($n > 3$) vertices, and let p be the perimeter. Prove that $\frac{n-3}{2} < \frac{d}{p} < \frac{1}{2} \left(\left\lceil \frac{n}{2} \right\rceil \left\lceil \frac{n+1}{2} \right\rceil - 2 \right)$.
6. Let a, b, c, d be odd positive integers such that $a < b < c < d$, $ad = bc$, and $a + d = 2^k$, $b + c = 2^m$ for some integers k, m . Prove that $a = 1$.

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